

DeepFake Detection

February 18, 2025

```
[2]: import dlib

print("All imports successful!")
```

All imports successful!

```
[4]: !pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in
c:\users\admin\anaconda3\envs\deepfake_detection\lib\site-packages (1.3.2)
Requirement already satisfied: numpy<2.0,>=1.17.3 in
c:\users\admin\anaconda3\envs\deepfake_detection\lib\site-packages (from scikit-
learn) (1.24.3)
Requirement already satisfied: scipy>=1.5.0 in
c:\users\admin\anaconda3\envs\deepfake_detection\lib\site-packages (from scikit-
learn) (1.10.1)
Requirement already satisfied: joblib>=1.1.1 in
c:\users\admin\anaconda3\envs\deepfake_detection\lib\site-packages (from scikit-
learn) (1.4.2)
Requirement already satisfied: threadpoolctl>=2.0.0 in
c:\users\admin\anaconda3\envs\deepfake_detection\lib\site-packages (from scikit-
learn) (3.5.0)
```

```
[6]: import os
import numpy as np
import matplotlib.pyplot as plt
import cv2

import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import Xception
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Dropout
from tensorflow.keras.models import Model
from tensorflow.keras.optimizers import Adam

from sklearn.metrics import confusion_matrix, classification_report

print("All libraries imported successfully!")
```

```
print("TensorFlow version:", tf.__version__)
```

All libraries imported successfully!

TensorFlow version: 2.13.0

```
[8]: import os
import cv2

# Define the base dataset directory
base_dir = r'C:\Users\Admin\Downloads\Compressed\FaceForensics++'

# Define video directories
real_video_dir = os.path.join(base_dir, 'real')
fake_video_dir = os.path.join(base_dir, 'fake')

# Define output directories for extracted frames (to be saved in the same main_
→directory)
extracted_real_dir = os.path.join(base_dir, 'extracted_real')
extracted_fake_dir = os.path.join(base_dir, 'extracted_fake')

# Create output directories if they don't exist
os.makedirs(extracted_real_dir, exist_ok=True)
os.makedirs(extracted_fake_dir, exist_ok=True)

# Function to extract frames from a video file
def extract_frames(video_path, output_dir, frame_interval=30):
    cap = cv2.VideoCapture(video_path)
    count = 0
    saved_count = 0
    while cap.isOpened():
        ret, frame = cap.read()
        if not ret:
            break
        if count % frame_interval == 0:
            filename = os.path.join(output_dir, f'{os.path.basename(video_path)}.
→split(".")[0]}_frame_{saved_count:05d}.jpg')
            cv2.imwrite(filename, frame)
            saved_count += 1
        count += 1
    cap.release()
    print(f"Extracted {saved_count} frames from {os.path.basename(video_path)}")

# Process videos in the "real" folder
for file in os.listdir(real_video_dir):
    if file.lower().endswith('.mp4'):
        video_file = os.path.join(real_video_dir, file)
        extract_frames(video_file, extracted_real_dir)
```

```

# Process videos in the "fake" folder
for file in os.listdir(fake_video_dir):
    if file.lower().endswith('.mp4'):
        video_file = os.path.join(fake_video_dir, file)
        extract_frames(video_file, extracted_fake_dir)

```

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Extracted 11 frames from 01__exit_phone_room.mp4
Extracted 27 frames from 01__hugging_happy.mp4
Extracted 19 frames from 01__kitchen_pan.mp4
Extracted 27 frames from 01__kitchen_still.mp4
Extracted 35 frames from 01__meeting_serious.mp4
Extracted 21 frames from 01__outside_talking_pan_laughing.mp4
Extracted 29 frames from 01__outside_talking_still_laughing.mp4
Extracted 31 frames from 01__podium_speech_happy.mp4
Extracted 33 frames from 01__secret_conversation.mp4
Extracted 29 frames from 01__talking_against_wall.mp4
Extracted 50 frames from 01__talking_angry_couch.mp4
Extracted 37 frames from 01__walking_and_outside_surprised.mp4
Extracted 28 frames from 01__walking_down_indoor_hall_disgust.mp4
Extracted 21 frames from 01__walking_down_street_outside_angry.mp4
Extracted 11 frames from 01__walking_outside_cafe_disgusted.mp4
Extracted 13 frames from 01__walk_down_hall_angry.mp4
Extracted 13 frames from 02__exit_phone_room.mp4
Extracted 27 frames from 02__hugging_happy.mp4
Extracted 20 frames from 02__kitchen_pan.mp4
Extracted 29 frames from 02__kitchen_still.mp4
Extracted 33 frames from 02__meeting_serious.mp4
Extracted 23 frames from 02__outside_talking_pan_laughing.mp4
Extracted 28 frames from 02__outside_talking_still_laughing.mp4
Extracted 36 frames from 02__podium_speech_happy.mp4
Extracted 34 frames from 02__secret_conversation.mp4
Extracted 33 frames from 02__talking_against_wall.mp4
Extracted 44 frames from 02__talking_angry_couch.mp4
Extracted 37 frames from 02__walking_and_outside_surprised.mp4
Extracted 28 frames from 02__walking_down_indoor_hall_disgust.mp4
Extracted 25 frames from 02__walking_down_street_outside_angry.mp4
Extracted 20 frames from 02__walking_outside_cafe_disgusted.mp4
Extracted 14 frames from 02__walk_down_hall_angry.mp4
Extracted 21 frames from 03__exit_phone_room.mp4
Extracted 28 frames from 03__hugging_happy.mp4
Extracted 19 frames from 03__kitchen_pan.mp4
Extracted 32 frames from 03__kitchen_still.mp4
Extracted 32 frames from 03__meeting_serious.mp4
Extracted 25 frames from 03__outside_talking_pan_laughing.mp4
Extracted 32 frames from 03__outside_talking_still_laughing.mp4
Extracted 28 frames from 03__podium_speech_happy.mp4
Extracted 29 frames from 03__secret_conversation.mp4

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Extracted 31 frames from 03__talking_against_wall.mp4
Extracted 49 frames from 03__talking_angry_couch.mp4
Extracted 39 frames from 03__walking_and_outside_surprised.mp4
Extracted 49 frames from 03__walking_down_indoor_hall_disgust.mp4
Extracted 41 frames from 03__walking_down_street_outside_angry.mp4
Extracted 24 frames from 03__walking_outside_cafe_disgusted.mp4
Extracted 20 frames from 03__walk_down_hall_angry.mp4
Extracted 13 frames from 04__exit_phone_room.mp4
Extracted 17 frames from 04__kitchen_pan.mp4
Extracted 28 frames from 04__kitchen_still.mp4
Extracted 27 frames from 04__outside_talking_pan_laughing.mp4
Extracted 27 frames from 04__outside_talking_still_laughing.mp4
Extracted 32 frames from 04__podium_speech_happy.mp4
Extracted 31 frames from 04__secret_conversation.mp4
Extracted 30 frames from 04__talking_against_wall.mp4
Extracted 46 frames from 04__talking_angry_couch.mp4
Extracted 20 frames from 04__walking_down_street_outside_angry.mp4
Extracted 14 frames from 04__walking_outside_cafe_disgusted.mp4
Extracted 12 frames from 04__walk_down_hall_angry.mp4
Extracted 18 frames from 05__exit_phone_room.mp4
Extracted 28 frames from 05__hugging_happy.mp4
Extracted 23 frames from 05__kitchen_pan.mp4
Extracted 30 frames from 05__kitchen_still.mp4
Extracted 26 frames from 05__outside_talking_pan_laughing.mp4
Extracted 32 frames from 05__outside_talking_still_laughing.mp4
Extracted 36 frames from 05__podium_speech_happy.mp4
Extracted 28 frames from 05__talking_against_wall.mp4
Extracted 29 frames from 05__walking_down_street_outside_angry.mp4
Extracted 22 frames from 05__walking_outside_cafe_disgusted.mp4
Extracted 20 frames from 05__walk_down_hall_angry.mp4
Extracted 10 frames from 06__exit_phone_room.mp4
Extracted 30 frames from 06__hugging_happy.mp4
Extracted 25 frames from 06__kitchen_pan.mp4
Extracted 31 frames from 06__kitchen_still.mp4
Extracted 23 frames from 06__outside_talking_pan_laughing.mp4
Extracted 29 frames from 06__outside_talking_still_laughing.mp4
Extracted 42 frames from 06__podium_speech_happy.mp4
Extracted 29 frames from 06__talking_against_wall.mp4
Extracted 40 frames from 06__talking_angry_couch.mp4
Extracted 34 frames from 06__walking_and_outside_surprised.mp4
Extracted 46 frames from 06__walking_down_indoor_hall_disgust.mp4
Extracted 26 frames from 06__walking_down_street_outside_angry.mp4
Extracted 12 frames from 06__walking_outside_cafe_disgusted.mp4
Extracted 15 frames from 06__walk_down_hall_angry.mp4
Extracted 17 frames from 07__exit_phone_room.mp4
Extracted 30 frames from 07__hugging_happy.mp4
Extracted 20 frames from 07__kitchen_pan.mp4
Extracted 28 frames from 07__kitchen_still.mp4

Extracted 24 frames from 07__outside_talking_pan_laughing.mp4
Extracted 29 frames from 07__outside_talking_still_laughing.mp4
Extracted 38 frames from 07__podium_speech_happy.mp4
Extracted 34 frames from 07__secret_conversation.mp4
Extracted 28 frames from 07__talking_against_wall.mp4
Extracted 54 frames from 07__talking_angry_couch.mp4
Extracted 27 frames from 07__walking_down_street_outside_angry.mp4
Extracted 19 frames from 07__walking_outside_cafe_disgusted.mp4
Extracted 25 frames from 07__walk_down_hall_angry.mp4
Extracted 13 frames from 08__exit_phone_room.mp4
Extracted 22 frames from 08__kitchen_pan.mp4
Extracted 34 frames from 08__kitchen_still.mp4
Extracted 24 frames from 08__outside_talking_pan_laughing.mp4
Extracted 30 frames from 08__outside_talking_still_laughing.mp4
Extracted 37 frames from 08__podium_speech_happy.mp4
Extracted 37 frames from 08__talking_against_wall.mp4
Extracted 24 frames from 08__walking_down_street_outside_angry.mp4
Extracted 14 frames from 08__walking_outside_cafe_disgusted.mp4
Extracted 16 frames from 08__walk_down_hall_angry.mp4
Extracted 15 frames from 09__exit_phone_room.mp4
Extracted 24 frames from 09__kitchen_pan.mp4
Extracted 28 frames from 09__outside_talking_pan_laughing.mp4
Extracted 32 frames from 09__outside_talking_still_laughing.mp4
Extracted 35 frames from 09__podium_speech_happy.mp4
Extracted 33 frames from 09__talking_against_wall.mp4
Extracted 45 frames from 09__talking_angry_couch.mp4
Extracted 19 frames from 09__walking_down_street_outside_angry.mp4
Extracted 16 frames from 09__walk_down_hall_angry.mp4
Extracted 11 frames from 10__exit_phone_room.mp4
Extracted 26 frames from 10__kitchen_pan.mp4
Extracted 30 frames from 10__kitchen_still.mp4
Extracted 27 frames from 10__outside_talking_pan_laughing.mp4
Extracted 27 frames from 10__outside_talking_still_laughing.mp4
Extracted 34 frames from 10__podium_speech_happy.mp4
Extracted 35 frames from 10__talking_against_wall.mp4
Extracted 48 frames from 10__talking_angry_couch.mp4
Extracted 34 frames from 10__walking_down_street_outside_angry.mp4
Extracted 35 frames from 10__walking_outside_cafe_disgusted.mp4
Extracted 30 frames from 10__walk_down_hall_angry.mp4
Extracted 28 frames from 11__exit_phone_room.mp4
Extracted 24 frames from 11__kitchen_pan.mp4
Extracted 27 frames from 11__kitchen_still.mp4
Extracted 25 frames from 11__outside_talking_pan_laughing.mp4
Extracted 27 frames from 11__outside_talking_still_laughing.mp4
Extracted 40 frames from 11__podium_speech_happy.mp4
Extracted 32 frames from 11__secret_conversation.mp4
Extracted 30 frames from 11__talking_against_wall.mp4
Extracted 54 frames from 11__talking_angry_couch.mp4

Extracted 37 frames from 11__walking_down_street_outside_angry.mp4
Extracted 25 frames from 11__walking_outside_cafe_disgusted.mp4
Extracted 29 frames from 11__walk_down_hall_angry.mp4
Extracted 30 frames from 12__exit_phone_room.mp4
Extracted 32 frames from 12__hugging_happy.mp4
Extracted 25 frames from 12__kitchen_pan.mp4
Extracted 28 frames from 12__kitchen_still.mp4
Extracted 31 frames from 12__outside_talking_pan_laughing.mp4
Extracted 29 frames from 12__outside_talking_still_laughing.mp4
Extracted 34 frames from 12__podium_speech_happy.mp4
Extracted 34 frames from 12__secret_conversation.mp4
Extracted 30 frames from 12__talking_against_wall.mp4
Extracted 47 frames from 12__talking_angry_couch.mp4
Extracted 32 frames from 12__walking_and_outside_surprised.mp4
Extracted 45 frames from 12__walking_down_indoor_hall_disgust.mp4
Extracted 24 frames from 12__walking_down_street_outside_angry.mp4
Extracted 26 frames from 12__walking_outside_cafe_disgusted.mp4
Extracted 18 frames from 12__walk_down_hall_angry.mp4
Extracted 18 frames from 13__exit_phone_room.mp4
Extracted 31 frames from 13__hugging_happy.mp4
Extracted 22 frames from 13__kitchen_pan.mp4
Extracted 26 frames from 13__kitchen_still.mp4
Extracted 23 frames from 13__outside_talking_pan_laughing.mp4
Extracted 30 frames from 13__outside_talking_still_laughing.mp4
Extracted 34 frames from 13__podium_speech_happy.mp4
Extracted 32 frames from 13__secret_conversation.mp4
Extracted 35 frames from 13__talking_against_wall.mp4
Extracted 45 frames from 13__talking_angry_couch.mp4
Extracted 32 frames from 13__walking_and_outside_surprised.mp4
Extracted 45 frames from 13__walking_down_indoor_hall_disgust.mp4
Extracted 25 frames from 13__walking_down_street_outside_angry.mp4
Extracted 13 frames from 13__walking_outside_cafe_disgusted.mp4
Extracted 21 frames from 13__walk_down_hall_angry.mp4
Extracted 22 frames from 14__exit_phone_room.mp4
Extracted 6 frames from 14__hugging_happy.mp4
Extracted 23 frames from 14__kitchen_pan.mp4
Extracted 30 frames from 14__kitchen_still.mp4
Extracted 25 frames from 14__outside_talking_pan_laughing.mp4
Extracted 31 frames from 14__outside_talking_still_laughing.mp4
Extracted 45 frames from 14__podium_speech_happy.mp4
Extracted 35 frames from 14__secret_conversation.mp4
Extracted 33 frames from 14__talking_against_wall.mp4
Extracted 54 frames from 14__talking_angry_couch.mp4
Extracted 29 frames from 14__walking_and_outside_surprised.mp4
Extracted 49 frames from 14__walking_down_indoor_hall_disgust.mp4
Extracted 49 frames from 14__walking_down_street_outside_angry.mp4
Extracted 25 frames from 14__walking_outside_cafe_disgusted.mp4
Extracted 26 frames from 14__walk_down_hall_angry.mp4

Extracted 15 frames from 15__exit_phone_room.mp4
Extracted 27 frames from 15__hugging_happy.mp4
Extracted 21 frames from 15__kitchen_pan.mp4
Extracted 29 frames from 15__kitchen_still.mp4
Extracted 24 frames from 15__outside_talking_pan_laughing.mp4
Extracted 29 frames from 15__outside_talking_still_laughing.mp4
Extracted 48 frames from 15__podium_speech_happy.mp4
Extracted 29 frames from 15__talking_against_wall.mp4
Extracted 45 frames from 15__talking_angry_couch.mp4
Extracted 30 frames from 15__walking_and_outside_surprised.mp4
Extracted 47 frames from 15__walking_down_indoor_hall_disgust.mp4
Extracted 18 frames from 15__walking_down_street_outside_angry.mp4
Extracted 19 frames from 15__walking_outside_cafe_disgusted.mp4
Extracted 19 frames from 15__walk_down_hall_angry.mp4
Extracted 20 frames from 16__exit_phone_room.mp4
Extracted 25 frames from 01_02__outside_talking_still_laughing__YVGY8LOK.mp4
Extracted 8 frames from 01_02__walk_down_hall_angry__YVGY8LOK.mp4
Extracted 21 frames from 01_03__hugging_happy__ISF9SP4G.mp4
Extracted 30 frames from 01_03__podium_speech_happy__480LQD1C.mp4
Extracted 29 frames from 01_03__talking_against_wall__JZUXXFRB.mp4
Extracted 35 frames from 01_11__meeting_serious__90M3VE0Y.mp4
Extracted 31 frames from 01_11__secret_conversation__40JNJL00.mp4
Extracted 29 frames from 01_11__talking_against_wall__9229VVZ3.mp4
Extracted 8 frames from 01_11__walking_outside_cafe_disgusted__FAFWDR4W.mp4
Extracted 10 frames from 01_12__outside_talking_pan_laughing__TNI7KUZ6.mp4
Extracted 17 frames from 01_20__outside_talking_pan_laughing__OTGHOG4Z.mp4
Extracted 25 frames from 01_20__outside_talking_still_laughing__FW94AIMJ.mp4
Extracted 8 frames from 01_21__walk_down_hall_angry__03X7CELV.mp4
Extracted 20 frames from 01_27__hugging_happy__ZYCZ30C0.mp4
Extracted 25 frames from 01_27__outside_talking_still_laughing__ZYCZ30C0.mp4
Extracted 8 frames from 01_27__walking_outside_cafe_disgusted__ZYCZ30C0.mp4
Extracted 22 frames from 02_01__secret_conversation__YVGY8LOK.mp4
Extracted 16 frames from 02_03__walking_down_street_outside_angry__QH3Y0IG0.mp4
Extracted 17 frames from 02_03__walking_outside_cafe_disgusted__QH3Y0IG0.mp4
Extracted 36 frames from 02_06__podium_speech_happy__N8OSN8P6.mp4
Extracted 43 frames from 02_06__talking_angry_couch__GH8TGTBS.mp4
Extracted 43 frames from 02_06__talking_angry_couch__MKZTXQ2T.mp4
Extracted 27 frames from 02_06__walking_and_outside_surprised__N8OSN8P6.mp4
Extracted 28 frames from 02_06__walking_down_indoor_hall_disgust__U6MDWIHG.mp4
Extracted 33 frames from 02_07__meeting_serious__1JCLEEBQ.mp4
Extracted 34 frames from 02_07__walking_and_outside_surprised__1VMZUH1W.mp4
Extracted 17 frames from 02_07__walking_down_street_outside_angry__04SXNLRL.mp4
Extracted 7 frames from 02_07__walk_down_hall_angry__U7DEOZNV.mp4
Extracted 9 frames from 02_09__exit_phone_room__HIH8YA82.mp4
Extracted 20 frames from 02_09__kitchen_pan__HIH8YA82.mp4
Extracted 36 frames from 02_12__podium_speech_happy__9D2ZHEKW.mp4
Extracted 9 frames from 02_13__exit_phone_room__CP5HFV3K.mp4
Extracted 22 frames from 02_13__outside_talking_pan_laughing__2YSYT2N3.mp4

Extracted 36 frames from 02_13__podium_speech_happy__2YSYT2N3.mp4
Extracted 2 frames from 02_13__secret_conversation__CP5HFV3K.mp4
Extracted 22 frames from 02_14__outside_talking_pan_laughing__3IUBEKCT.mp4
Extracted 9 frames from 02_15__exit_phone_room__I8G2LWD1.mp4
Extracted 9 frames from 02_15__exit_phone_room__MZWH8ATN.mp4
Extracted 22 frames from 02_15__outside_talking_pan_laughing__I8G2LWD1.mp4
Extracted 21 frames from 02_15__outside_talking_pan_laughing__SB6PMC00.mp4
Extracted 2 frames from 02_15__secret_conversation__MZWH8ATN.mp4
Extracted 33 frames from 02_15__talking_against_wall__HTG660F8.mp4
Extracted 43 frames from 02_15__talking_angry_couch__HTG660F8.mp4
Extracted 43 frames from 02_15__talking_angry_couch__I8G2LWD1.mp4
Extracted 34 frames from 02_15__walking_and_outside_surprised__HTG660F8.mp4
Extracted 33 frames from 02_15__walking_and_outside_surprised__I8G2LWD1.mp4
Extracted 26 frames from 02_15__walking_and_outside_surprised__MZWH8ATN.mp4
Extracted 7 frames from 02_15__walk_down_hall_angry__TN2CWM3K.mp4
Extracted 9 frames from 02_18__exit_phone_room__OXMEEFUQ.mp4
Extracted 22 frames from 02_18__outside_talking_pan_laughing__OXMEEFUQ.mp4
Extracted 16 frames from 02_18__walking_down_street_outside_angry__21JTDEL.mp4
Extracted 20 frames from 02_21__kitchen_pan__ZOXHPQAR.mp4
Extracted 43 frames from 02_21__talking_angry_couch__ZOXHPQAR.mp4
Extracted 33 frames from 02_25__talking_against_wall__Z7FQ69VP.mp4
Extracted 23 frames from 02_27__hugging_happy__GVFLSZD5.mp4
Extracted 7 frames from 02_27__walk_down_hall_angry__78M8S6M6.mp4
Extracted 37 frames from 03_01__walking_and_outside_surprised__JZUXXFRB.mp4
Extracted 24 frames from 03_04__outside_talking_pan_laughing__T04P6ELC.mp4
Extracted 49 frames from 03_04__talking_angry_couch__T04P6ELC.mp4
Extracted 32 frames from 03_06__meeting_serious__83ABVHC3.mp4
Extracted 28 frames from 03_06__podium_speech_happy__83ABVHC3.mp4
Extracted 28 frames from 03_07__hugging_happy__BKLOCI1M.mp4
Extracted 33 frames from 03_07__walking_down_indoor_hall_disgust__PWXXULHR.mp4
Extracted 22 frames from 03_07__walking_outside_cafe_disgusted__CDSNLDQ8.mp4
Extracted 23 frames from 03_07__walking_outside_cafe_disgusted__IFSURI9X.mp4
Extracted 24 frames from 03_07__walking_outside_cafe_disgusted__PWXXULHR.mp4
Extracted 15 frames from 03_07__walk_down_hall_angry__6PHZRQ4H.mp4
Extracted 15 frames from 03_07__walk_down_hall_angry__IFSURI9X.mp4
Extracted 32 frames from 03_09__outside_talking_still_laughing__RCETIXYL.mp4
Extracted 27 frames from 03_09__secret_conversation__RCETIXYL.mp4
Extracted 18 frames from 03_11__exit_phone_room__P08VGHTA.mp4
Extracted 31 frames from 03_11__talking_against_wall__P08VGHTA.mp4
Extracted 18 frames from 03_13__kitchen_pan__GBYWJW06.mp4
Extracted 32 frames from 03_13__meeting_serious__GBYWJW06.mp4
Extracted 32 frames from 03_13__meeting_serious__T3MZOI8X.mp4
Extracted 32 frames from 03_13__outside_talking_still_laughing__GBYWJW06.mp4
Extracted 28 frames from 03_14__podium_speech_happy__Q9NSXM88.mp4
Extracted 31 frames from 03_14__talking_against_wall__ZC2KYASW.mp4
Extracted 32 frames from 03_14__walking_down_street_outside_angry__HOVQHGS3.mp4
Extracted 16 frames from 03_14__walk_down_hall_angry__P1L5PF4I.mp4
Extracted 18 frames from 03_15__kitchen_pan__AIOM1U5V.mp4

Extracted 18 frames from 03_15__kitchen_pan__DNUJD8M2.mp4
Extracted 24 frames from 03_15__outside_talking_pan_laughing__Y11NT1YX.mp4
Extracted 32 frames from 03_15__outside_talking_still_laughing__DNUJD8M2.mp4
Extracted 28 frames from 03_15__podium_speech_happy__DG8ITQ03.mp4
Extracted 26 frames from 03_18__walking_and_outside_surprised__8HNSXOBW.mp4
Extracted 24 frames from 03_18__walking_outside_cafe_disgusted__22UBCOBS.mp4
Extracted 19 frames from 03_21__exit_phone_room__YCSEBZ04.mp4
Extracted 18 frames from 03_21__kitchen_pan__YCSEBZ04.mp4
Extracted 32 frames from 03_21__meeting_serious__V53E3RVB.mp4
Extracted 27 frames from 03_21__secret_conversation__YCSEBZ04.mp4
Extracted 18 frames from 03_26__kitchen_pan__WBSXBL82.mp4
Extracted 32 frames from 03_26__kitchen_still__WBSXBL82.mp4
Extracted 31 frames from 03_26__talking_against_wall__WBSXBL82.mp4
Extracted 49 frames from 03_27__walking_down_indoor_hall_disgust__IL675GCI.mp4
Extracted 15 frames from 03_27__walk_down_hall_angry__IL675GCI.mp4
Extracted 13 frames from 04_01__exit_phone_room__OXUW13RW.mp4
Extracted 28 frames from 04_01__kitchen_still__6I623VU9.mp4
Extracted 27 frames from 04_01__outside_talking_still_laughing__OXUW13RW.mp4
Extracted 27 frames from 04_01__outside_talking_still_laughing__6I623VU9.mp4
Extracted 29 frames from 04_01__secret_conversation__6I623VU9.mp4
Extracted 28 frames from 04_06__kitchen_still__ZK95PQDE.mp4
Extracted 11 frames from 04_07__exit_phone_room__ITCOC48B.mp4
Extracted 17 frames from 04_07__kitchen_pan__XRK7FGZX.mp4
Extracted 16 frames from 04_12__walking_down_street_outside_angry__96TQKDFJ.mp4
Extracted 32 frames from 04_13__podium_speech_happy__00T3UYOR.mp4
Extracted 29 frames from 04_13__secret_conversation__00T3UYOR.mp4
Extracted 15 frames from 04_13__walking_down_street_outside_angry__00T3UYOR.mp4
Extracted 13 frames from 04_13__walking_outside_cafe_disgusted__00T3UYOR.mp4
Extracted 28 frames from 04_18__kitchen_still__NAXINA1N.mp4
Extracted 16 frames from 04_21__walking_down_street_outside_angry__5Y31RZP8.mp4
Extracted 27 frames from 04_26__outside_talking_still_laughing__WX836VLY.mp4
Extracted 28 frames from 05_08__talking_against_wall__PRBCE28Z.mp4
Extracted 14 frames from 05_16__exit_phone_room__B62WCGUN.mp4
Extracted 14 frames from 05_16__walk_down_hall_angry__OSXCUOHX.mp4
Extracted 14 frames from 05_16__walk_down_hall_angry__U9WZI5LK.mp4
Extracted 28 frames from 05_17__hugging_happy__YTJYYD09.mp4
Extracted 22 frames from 05_17__walking_down_street_outside_angry__M3H96PDQ.mp4
Extracted 14 frames from 05_28__exit_phone_room__U9LRLJ6N.mp4
Extracted 30 frames from 05_28__kitchen_still__W3J028UG.mp4
Extracted 42 frames from 06_02__podium_speech_happy__N80SN8P6.mp4
Extracted 11 frames from 06_02__walk_down_hall_angry__37DH75GQ.mp4
Extracted 29 frames from 06_03__talking_against_wall__4I8LRXWF.mp4
Extracted 23 frames from 06_04__outside_talking_still_laughing__ZK95PQDE.mp4
Extracted 11 frames from 06_04__walking_outside_cafe_disgusted__ZK95PQDE.mp4
Extracted 9 frames from 06_07__exit_phone_room__NMGYPBXE.mp4
Extracted 23 frames from 06_07__kitchen_pan__NMGYPBXE.mp4
Extracted 19 frames from 06_07__walking_down_street_outside_angry__NMGYPBXE.mp4
Extracted 11 frames from 06_11__walking_outside_cafe_disgusted__MX659QU8.mp4

Extracted 20 frames from 06_12__outside_talking_pan_laughing__3K21NFMN.mp4
Extracted 42 frames from 06_12__podium_speech_happy__OVR4Y891.mp4
Extracted 42 frames from 06_12__podium_speech_happy__3K21NFMN.mp4
Extracted 21 frames from 06_14__outside_talking_pan_laughing__8U9ULZDT.mp4
Extracted 33 frames from 06_14__walking_and_outside_surprised__8U9ULZDT.mp4
Extracted 46 frames from 06_14__walking_down_indoor_hall_disgust__8U9ULZDT.mp4
Extracted 9 frames from 06_15__exit_phone_room__QRCD27P8.mp4
Extracted 24 frames from 06_15__outside_talking_still_laughing__QRCD27P8.mp4
Extracted 33 frames from 06_15__walking_and_outside_surprised__QRCD27P8.mp4
Extracted 21 frames from 06_18__outside_talking_pan_laughing__DEA1TCLN.mp4
Extracted 23 frames from 06_18__outside_talking_still_laughing__M36D00JT.mp4
Extracted 42 frames from 06_18__podium_speech_happy__DEA1TCLN.mp4
Extracted 11 frames from 06_18__walk_down_hall_angry__LHOKWJKM.mp4
Extracted 23 frames from 06_20__kitchen_pan__6SUW7063.mp4
Extracted 20 frames from 06_25__outside_talking_pan_laughing__MI9BDQ7M.mp4
Extracted 40 frames from 06_25__talking_angry_couch__MI9BDQ7M.mp4
Extracted 33 frames from 06_25__walking_and_outside_surprised__MI9BDQ7M.mp4
Extracted 23 frames from 06_26__kitchen_pan__L5BVR5L9.mp4
Extracted 31 frames from 06_26__kitchen_still__L5BVR5L9.mp4
Extracted 40 frames from 06_27__talking_angry_couch__JOG5PB18.mp4
Extracted 34 frames from 06_27__walking_and_outside_surprised__07L5Z9U8.mp4
Extracted 53 frames from 07_02__talking_angry_couch__1H07DFQJ.mp4
Extracted 20 frames from 07_02__walking_down_street_outside_angry__04SXLRL.mp4
Extracted 25 frames from 07_03__hugging_happy__7NGMD8FT.mp4
Extracted 24 frames from 07_03__outside_talking_pan_laughing__IFSURI9X.mp4
Extracted 36 frames from 07_03__podium_speech_happy__6PHZRQ4H.mp4
Extracted 23 frames from 07_03__secret_conversation__IFSURI9X.mp4
Extracted 52 frames from 07_03__talking_angry_couch__WPT3Z2KN.mp4
Extracted 16 frames from 07_03__walking_outside_cafe_disgusted__F0YYEA5W.mp4
Extracted 18 frames from 07_06__kitchen_pan__NMGYPBXE.mp4
Extracted 29 frames from 07_06__outside_talking_still_laughing__NMGYPBXE.mp4
Extracted 19 frames from 07_06__walking_down_street_outside_angry__NMGYPBXE.mp4
Extracted 29 frames from 07_09__outside_talking_still_laughing__N9CWME71.mp4
Extracted 19 frames from 07_09__walk_down_hall_angry__N9CWME71.mp4
Extracted 21 frames from 07_12__outside_talking_still_laughing__OVN0A2T3.mp4
Extracted 15 frames from 07_13__walking_outside_cafe_disgusted__RVQCPCJF.mp4
Extracted 14 frames from 07_14__exit_phone_room__P9QF050U.mp4
Extracted 26 frames from 07_14__talking_against_wall__P9QF050U.mp4
Extracted 25 frames from 07_15__hugging_happy__9Z2MLKVX.mp4
Extracted 24 frames from 07_20__outside_talking_pan_laughing__KV6Q7D6C.mp4
Extracted 25 frames from 07_20__talking_against_wall__KV6Q7D6C.mp4
Extracted 14 frames from 07_21__exit_phone_room__K7KXUHMU.mp4
Extracted 28 frames from 07_21__outside_talking_still_laughing__K7KXUHMU.mp4
Extracted 22 frames from 07_21__talking_against_wall__MKU99DVX.mp4
Extracted 15 frames from 07_25__walk_down_hall_angry__PAE9HCA8.mp4
Extracted 19 frames from 07_26__walking_down_street_outside_angry__FGNGC2GT.mp4
Extracted 19 frames from 07_26__walk_down_hall_angry__FGNGC2GT.mp4
Extracted 20 frames from 07_27__walking_down_street_outside_angry__3RH5PR6S.mp4

Extracted 10 frames from 08_05__walk_down_hall_angry__FBICSP2C.mp4
 Extracted 36 frames from 08_16__podium_speech_happy__8Q7JCS95.mp4
 Extracted 20 frames from 08_28__outside_talking_pan_laughing__8BC35RFU.mp4
 Extracted 33 frames from 09_01__talking_against_wall__08HNNX43.mp4
 Extracted 12 frames from 09_01__walk_down_hall_angry__6TSGVLHA.mp4
 Extracted 13 frames from 09_02__walking_down_street_outside_angry__6KUOFMZW.mp4
 Extracted 15 frames from 09_02__walking_down_street_outside_angry__9TDCEK1Q.mp4
 Extracted 11 frames from 09_02__walk_down_hall_angry__6KUOFMZW.mp4
 Extracted 23 frames from 09_03__kitchen_pan__8DTEGQ54.mp4
 Extracted 33 frames from 09_03__talking_against_wall__8DTEGQ54.mp4
 Extracted 22 frames from 09_07__kitchen_pan__N9CWME71.mp4
 Extracted 22 frames from 09_13__kitchen_pan__21H6XSPE.mp4
 Extracted 24 frames from 09_13__outside_talking_pan_laughing__LPT427RY.mp4
 Extracted 25 frames from 09_18__outside_talking_pan_laughing__3VP8836C.mp4
 Extracted 22 frames from 09_20__kitchen_pan__98NUQ3E6.mp4
 Extracted 35 frames from 09_20__podium_speech_happy__05X0AWR9.mp4
 Extracted 45 frames from 09_20__talking_angry_couch__0IRM5ADD.mp4
 Extracted 14 frames from 09_25__walking_down_street_outside_angry__50410DBN.mp4
 Extracted 33 frames from 09_26__talking_against_wall__C3K20JOL.mp4
 Extracted 12 frames from 09_26__walk_down_hall_angry__QSE5A0GF.mp4
 Extracted 30 frames from 10_19__kitchen_still__IDX76N5R.mp4
 Extracted 23 frames from 10_22__kitchen_pan__EHARPYBT.mp4
 Extracted 29 frames from 10_22__kitchen_still__EHARPYBT.mp4

```

[10]: import os
import shutil

# Define the source directories (where the extracted frames are currently stored)
source_real_dir = os.path.join(base_dir, 'extracted_real')
source_fake_dir = os.path.join(base_dir, 'extracted_fake')

# Define the destination directories (flat structure)
flat_data_dir = os.path.join(base_dir, 'data_flat')
flat_real_dir = os.path.join(flat_data_dir, 'real')
flat_fake_dir = os.path.join(flat_data_dir, 'fake')

# Create the flat directories if they don't exist
os.makedirs(flat_real_dir, exist_ok=True)
os.makedirs(flat_fake_dir, exist_ok=True)

# Function to copy images from nested folders to a flat folder
def flatten_folder(source_dir, dest_dir):
    for root, dirs, files in os.walk(source_dir):
        for file in files:
            if file.lower().endswith(('.jpg', '.png')):
                src_file = os.path.join(root, file)
                dest_file = os.path.join(dest_dir, file)
  
```

```

        # In case of duplicate file names, you can add a unique prefix
        ↪or suffix
        if os.path.exists(dest_file):
            dest_file = os.path.join(dest_dir, f"{os.path.
            ↪basename(root)}_{file}")
            shutil.copy(src_file, dest_file)

# Flatten the "real" images
flatten_folder(source_real_dir, flat_real_dir)
# Flatten the "fake" images
flatten_folder(source_fake_dir, flat_fake_dir)

print("Flattening complete. Check the 'data_flat' folder in your FaceForensics++
    ↪directory.")

```

Flattening complete. Check the 'data_flat' folder in your FaceForensics++ directory.

```

[12]: from tensorflow.keras.preprocessing.image import ImageDataGenerator

# Set the path to the flat dataset directory
data_flat_dir = os.path.join(base_dir, 'data_flat')

# Create an ImageDataGenerator with augmentation and a validation split.
datagen = ImageDataGenerator(
    rescale=1./255,          # Normalize pixel values to [0, 1]
    rotation_range=20,       # Random rotations for augmentation
    width_shift_range=0.2,   # Random horizontal shifts
    height_shift_range=0.2,  # Random vertical shifts
    horizontal_flip=True,    # Random horizontal flips
    validation_split=0.2     # Reserve 20% of data for validation
)

# Create the training data generator
train_generator = datagen.flow_from_directory(
    directory=data_flat_dir,
    target_size=(299, 299),  # Xception expects 299x299 input images
    batch_size=32,
    class_mode='binary',    # Binary classification: real vs fake
    subset='training',      # Use training split (80% of data)
    shuffle=True
)

# Create the validation data generator
validation_generator = datagen.flow_from_directory(
    directory=data_flat_dir,
    target_size=(299, 299),

```

```

        batch_size=32,
        class_mode='binary',
        subset='validation',          # Use validation split (20% of data)
        shuffle=False
    )

    print("Data generators created successfully!")

```

Found 8453 images belonging to 2 classes.

Found 2112 images belonging to 2 classes.

Data generators created successfully!

```

[14]: from tensorflow.keras.applications import Xception
      from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Dropout
      from tensorflow.keras.models import Model
      from tensorflow.keras.optimizers import Adam

      # Load the pre-trained Xception model without the top layers
      base_model = Xception(weights='imagenet', include_top=False, input_shape=(299,
      ↪299, 3))

      # Freeze the base model layers so they don't get re-trained initially
      for layer in base_model.layers:
          layer.trainable = False

      # Add custom layers on top of the base model
      x = base_model.output
      x = GlobalAveragePooling2D()(x)  # Convert the 2D feature maps into a single 1D
      ↪vector
      x = Dense(128, activation='relu')(x)  # Add a fully-connected layer with 128
      ↪units
      x = Dropout(0.5)(x)  # Add dropout for regularization (prevents overfitting)
      predictions = Dense(1, activation='sigmoid')(x)  # Final output layer for binary
      ↪classification

      # Define the complete model
      model = Model(inputs=base_model.input, outputs=predictions)

      # Compile the model
      model.compile(optimizer=Adam(learning_rate=1e-4),
                    loss='binary_crossentropy',
                    metrics=['accuracy'])

      # Print the model summary
      model.summary()

```

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/xception/xception_weights_tf_dim_ordering_tf_kernels_notop.h5

83683744/83683744 [=====] - 27s 0us/step

Model: "model"

Layer (type)	Output Shape	Param #	Connected to
input_1 (InputLayer)	[(None, 299, 299, 3)]	0	[]
block1_conv1 (Conv2D) ['input_1[0][0]']	(None, 149, 149, 32)	864	
block1_conv1_bn (BatchNorm ['block1_conv1[0][0]'] alization)	(None, 149, 149, 32)	128	
block1_conv1_act (Activati ['block1_conv1_bn[0][0]'] on)	(None, 149, 149, 32)	0	
block1_conv2 (Conv2D) ['block1_conv1_act[0][0]']	(None, 147, 147, 64)	18432	
block1_conv2_bn (BatchNorm ['block1_conv2[0][0]'] alization)	(None, 147, 147, 64)	256	
block1_conv2_act (Activati ['block1_conv2_bn[0][0]'] on)	(None, 147, 147, 64)	0	
block2_sepconv1 (Separable ['block1_conv2_act[0][0]'] Conv2D)	(None, 147, 147, 128)	8768	
block2_sepconv1_bn (BatchN ['block2_sepconv1[0][0]'] ormalization)	(None, 147, 147, 128)	512	
block2_sepconv2_act (Activ ['block2_sepconv1_bn[0][0]'] ation)	(None, 147, 147, 128)	0	
block2_sepconv2 (Separable ['block2_sepconv2_act[0][0]'] Conv2D)	(None, 147, 147, 128)	17536	
block2_sepconv2_bn (BatchN	(None, 147, 147, 128)	512	

['block2_sepconv2[0][0]'] ormalization)		
conv2d (Conv2D) ['block1_conv2_act[0][0]']	(None, 74, 74, 128)	8192
block2_pool (MaxPooling2D) ['block2_sepconv2_bn[0][0]']	(None, 74, 74, 128)	0
batch_normalization (Batch ['conv2d[0][0]'] Normalization)	(None, 74, 74, 128)	512
add (Add) ['block2_pool[0][0]', 'batch_normalization[0][0]']	(None, 74, 74, 128)	0
block3_sepconv1_act (Activ ['add[0][0]'] ation)	(None, 74, 74, 128)	0
block3_sepconv1 (Separable ['block3_sepconv1_act[0][0]'] Conv2D)	(None, 74, 74, 256)	33920
block3_sepconv1_bn (BatchN ['block3_sepconv1[0][0]'] ormalization)	(None, 74, 74, 256)	1024
block3_sepconv2_act (Activ ['block3_sepconv1_bn[0][0]'] ation)	(None, 74, 74, 256)	0
block3_sepconv2 (Separable ['block3_sepconv2_act[0][0]'] Conv2D)	(None, 74, 74, 256)	67840
block3_sepconv2_bn (BatchN ['block3_sepconv2[0][0]'] ormalization)	(None, 74, 74, 256)	1024
conv2d_1 (Conv2D) ['add[0][0]']	(None, 37, 37, 256)	32768
block3_pool (MaxPooling2D) ['block3_sepconv2_bn[0][0]']	(None, 37, 37, 256)	0
batch_normalization_1 (Bat	(None, 37, 37, 256)	1024

```

['conv2d_1[0][0]']
chNormalization)

add_1 (Add) (None, 37, 37, 256) 0
['block3_pool[0][0]',
'batch_normalization_1[0][0]']
]

block4_sepconv1_act (Activ (None, 37, 37, 256) 0
['add_1[0][0]']
ation)

block4_sepconv1 (Separable (None, 37, 37, 728) 188672
['block4_sepconv1_act[0][0]']
Conv2D)

block4_sepconv1_bn (BatchN (None, 37, 37, 728) 2912
['block4_sepconv1[0][0]']
ormalization)

block4_sepconv2_act (Activ (None, 37, 37, 728) 0
['block4_sepconv1_bn[0][0]']
ation)

block4_sepconv2 (Separable (None, 37, 37, 728) 536536
['block4_sepconv2_act[0][0]']
Conv2D)

block4_sepconv2_bn (BatchN (None, 37, 37, 728) 2912
['block4_sepconv2[0][0]']
ormalization)

conv2d_2 (Conv2D) (None, 19, 19, 728) 186368
['add_1[0][0]']

block4_pool (MaxPooling2D) (None, 19, 19, 728) 0
['block4_sepconv2_bn[0][0]']

batch_normalization_2 (Bat (None, 19, 19, 728) 2912
['conv2d_2[0][0]']
chNormalization)

add_2 (Add) (None, 19, 19, 728) 0
['block4_pool[0][0]',
'batch_normalization_2[0][0]']
]

block5_sepconv1_act (Activ (None, 19, 19, 728) 0

```

['add_2[0][0]'] ation)	
block5_sepconv1 (Separable (None, 19, 19, 728) ['block5_sepconv1_act[0][0]'] Conv2D)	536536
block5_sepconv1_bn (BatchN (None, 19, 19, 728) ['block5_sepconv1[0][0]'] ormalization)	2912
block5_sepconv2_act (Activ (None, 19, 19, 728) ['block5_sepconv1_bn[0][0]'] ation)	0
block5_sepconv2 (Separable (None, 19, 19, 728) ['block5_sepconv2_act[0][0]'] Conv2D)	536536
block5_sepconv2_bn (BatchN (None, 19, 19, 728) ['block5_sepconv2[0][0]'] ormalization)	2912
block5_sepconv3_act (Activ (None, 19, 19, 728) ['block5_sepconv2_bn[0][0]'] ation)	0
block5_sepconv3 (Separable (None, 19, 19, 728) ['block5_sepconv3_act[0][0]'] Conv2D)	536536
block5_sepconv3_bn (BatchN (None, 19, 19, 728) ['block5_sepconv3[0][0]'] ormalization)	2912
add_3 (Add) (None, 19, 19, 728) ['block5_sepconv3_bn[0][0]', 'add_2[0][0]']	0
block6_sepconv1_act (Activ (None, 19, 19, 728) ['add_3[0][0]'] ation)	0
block6_sepconv1 (Separable (None, 19, 19, 728) ['block6_sepconv1_act[0][0]'] Conv2D)	536536
block6_sepconv1_bn (BatchN (None, 19, 19, 728)	2912

['block6_sepconv1[0][0]'] ormalization)	
block6_sepconv2_act (Activ (None, 19, 19, 728) ['block6_sepconv1_bn[0][0]'] ation)	0
block6_sepconv2 (Separable (None, 19, 19, 728) ['block6_sepconv2_act[0][0]'] Conv2D)	536536
block6_sepconv2_bn (BatchN (None, 19, 19, 728) ['block6_sepconv2[0][0]'] ormalization)	2912
block6_sepconv3_act (Activ (None, 19, 19, 728) ['block6_sepconv2_bn[0][0]'] ation)	0
block6_sepconv3 (Separable (None, 19, 19, 728) ['block6_sepconv3_act[0][0]'] Conv2D)	536536
block6_sepconv3_bn (BatchN (None, 19, 19, 728) ['block6_sepconv3[0][0]'] ormalization)	2912
add_4 (Add) (None, 19, 19, 728) ['block6_sepconv3_bn[0][0]', 'add_3[0][0]']	0
block7_sepconv1_act (Activ (None, 19, 19, 728) ['add_4[0][0]'] ation)	0
block7_sepconv1 (Separable (None, 19, 19, 728) ['block7_sepconv1_act[0][0]'] Conv2D)	536536
block7_sepconv1_bn (BatchN (None, 19, 19, 728) ['block7_sepconv1[0][0]'] ormalization)	2912
block7_sepconv2_act (Activ (None, 19, 19, 728) ['block7_sepconv1_bn[0][0]'] ation)	0
block7_sepconv2 (Separable (None, 19, 19, 728)	536536

['block7_sepconv2_act[0][0]'] Conv2D)	
block7_sepconv2_bn (BatchN (None, 19, 19, 728) ['block7_sepconv2[0][0]'] ormalization)	2912
block7_sepconv3_act (Activ (None, 19, 19, 728) ['block7_sepconv2_bn[0][0]'] ation)	0
block7_sepconv3 (Separable (None, 19, 19, 728) ['block7_sepconv3_act[0][0]'] Conv2D)	536536
block7_sepconv3_bn (BatchN (None, 19, 19, 728) ['block7_sepconv3[0][0]'] ormalization)	2912
add_5 (Add) (None, 19, 19, 728) ['block7_sepconv3_bn[0][0]'], 'add_4[0][0]']	0
block8_sepconv1_act (Activ (None, 19, 19, 728) ['add_5[0][0]'] ation)	0
block8_sepconv1 (Separable (None, 19, 19, 728) ['block8_sepconv1_act[0][0]'] Conv2D)	536536
block8_sepconv1_bn (BatchN (None, 19, 19, 728) ['block8_sepconv1[0][0]'] ormalization)	2912
block8_sepconv2_act (Activ (None, 19, 19, 728) ['block8_sepconv1_bn[0][0]'] ation)	0
block8_sepconv2 (Separable (None, 19, 19, 728) ['block8_sepconv2_act[0][0]'] Conv2D)	536536
block8_sepconv2_bn (BatchN (None, 19, 19, 728) ['block8_sepconv2[0][0]'] ormalization)	2912
block8_sepconv3_act (Activ (None, 19, 19, 728)	0

['block8_sepconv2_bn[0][0]'] ation)	
block8_sepconv3 (Separable (None, 19, 19, 728) ['block8_sepconv3_act[0][0]'] Conv2D)	536536
block8_sepconv3_bn (BatchN (None, 19, 19, 728) ['block8_sepconv3[0][0]'] ormalization)	2912
add_6 (Add) (None, 19, 19, 728) ['block8_sepconv3_bn[0][0]'], 'add_5[0][0]']	0
block9_sepconv1_act (Activ (None, 19, 19, 728) ['add_6[0][0]'] ation)	0
block9_sepconv1 (Separable (None, 19, 19, 728) ['block9_sepconv1_act[0][0]'] Conv2D)	536536
block9_sepconv1_bn (BatchN (None, 19, 19, 728) ['block9_sepconv1[0][0]'] ormalization)	2912
block9_sepconv2_act (Activ (None, 19, 19, 728) ['block9_sepconv1_bn[0][0]'] ation)	0
block9_sepconv2 (Separable (None, 19, 19, 728) ['block9_sepconv2_act[0][0]'] Conv2D)	536536
block9_sepconv2_bn (BatchN (None, 19, 19, 728) ['block9_sepconv2[0][0]'] ormalization)	2912
block9_sepconv3_act (Activ (None, 19, 19, 728) ['block9_sepconv2_bn[0][0]'] ation)	0
block9_sepconv3 (Separable (None, 19, 19, 728) ['block9_sepconv3_act[0][0]'] Conv2D)	536536
block9_sepconv3_bn (BatchN (None, 19, 19, 728)	2912

['block9_sepconv3[0][0]'] ormalization)		
add_7 (Add) ['block9_sepconv3_bn[0][0]', 'add_6[0][0]']	(None, 19, 19, 728)	0
block10_sepconv1_act (Acti ['add_7[0][0]'] vation)	(None, 19, 19, 728)	0
block10_sepconv1 (Separabl ['block10_sepconv1_act[0][0]'] eConv2D)	(None, 19, 19, 728)	536536
block10_sepconv1_bn (Batch ['block10_sepconv1[0][0]'] Normalization)	(None, 19, 19, 728)	2912
block10_sepconv2_act (Acti ['block10_sepconv1_bn[0][0]'] vation)	(None, 19, 19, 728)	0
block10_sepconv2 (Separabl ['block10_sepconv2_act[0][0]'] eConv2D)	(None, 19, 19, 728)	536536
block10_sepconv2_bn (Batch ['block10_sepconv2[0][0]'] Normalization)	(None, 19, 19, 728)	2912
block10_sepconv3_act (Acti ['block10_sepconv2_bn[0][0]'] vation)	(None, 19, 19, 728)	0
block10_sepconv3 (Separabl ['block10_sepconv3_act[0][0]'] eConv2D)	(None, 19, 19, 728)	536536
block10_sepconv3_bn (Batch ['block10_sepconv3[0][0]'] Normalization)	(None, 19, 19, 728)	2912
add_8 (Add) ['block10_sepconv3_bn[0][0]', 'add_7[0][0]']	(None, 19, 19, 728)	0
block11_sepconv1_act (Acti	(None, 19, 19, 728)	0

['add_8[0][0]'] vation)	
block11_sepconv1 (Separabl (None, 19, 19, 728) ['block11_sepconv1_act[0][0]'] eConv2D)	536536
block11_sepconv1_bn (Batch (None, 19, 19, 728) ['block11_sepconv1[0][0]'] Normalization)	2912
block11_sepconv2_act (Acti (None, 19, 19, 728) ['block11_sepconv1_bn[0][0]'] vation)	0
block11_sepconv2 (Separabl (None, 19, 19, 728) ['block11_sepconv2_act[0][0]'] eConv2D)	536536
block11_sepconv2_bn (Batch (None, 19, 19, 728) ['block11_sepconv2[0][0]'] Normalization)	2912
block11_sepconv3_act (Acti (None, 19, 19, 728) ['block11_sepconv2_bn[0][0]'] vation)	0
block11_sepconv3 (Separabl (None, 19, 19, 728) ['block11_sepconv3_act[0][0]'] eConv2D)	536536
block11_sepconv3_bn (Batch (None, 19, 19, 728) ['block11_sepconv3[0][0]'] Normalization)	2912
add_9 (Add) (None, 19, 19, 728) ['block11_sepconv3_bn[0][0]', 'add_8[0][0]']	0
block12_sepconv1_act (Acti (None, 19, 19, 728) ['add_9[0][0]'] vation)	0
block12_sepconv1 (Separabl (None, 19, 19, 728) ['block12_sepconv1_act[0][0]'] eConv2D)	536536
block12_sepconv1_bn (Batch (None, 19, 19, 728)	2912

['block12_sepconv1[0][0]'] Normalization)	
block12_sepconv2_act (Acti (None, 19, 19, 728) ['block12_sepconv1_bn[0][0]'] vation)	0
block12_sepconv2 (Separabl (None, 19, 19, 728) ['block12_sepconv2_act[0][0]'] eConv2D)	536536
block12_sepconv2_bn (Batch (None, 19, 19, 728) ['block12_sepconv2[0][0]'] Normalization)	2912
block12_sepconv3_act (Acti (None, 19, 19, 728) ['block12_sepconv2_bn[0][0]'] vation)	0
block12_sepconv3 (Separabl (None, 19, 19, 728) ['block12_sepconv3_act[0][0]'] eConv2D)	536536
block12_sepconv3_bn (Batch (None, 19, 19, 728) ['block12_sepconv3[0][0]'] Normalization)	2912
add_10 (Add) (None, 19, 19, 728) ['block12_sepconv3_bn[0][0]', 'add_9[0][0]']	0
block13_sepconv1_act (Acti (None, 19, 19, 728) ['add_10[0][0]'] vation)	0
block13_sepconv1 (Separabl (None, 19, 19, 728) ['block13_sepconv1_act[0][0]'] eConv2D)	536536
block13_sepconv1_bn (Batch (None, 19, 19, 728) ['block13_sepconv1[0][0]'] Normalization)	2912
block13_sepconv2_act (Acti (None, 19, 19, 728) ['block13_sepconv1_bn[0][0]'] vation)	0
block13_sepconv2 (Separabl (None, 19, 19, 1024)	752024

```

['block13_sepconv2_act[0][0]']
    eConv2D)

    block13_sepconv2_bn (Batch (None, 19, 19, 1024)      4096
['block13_sepconv2[0][0]']
    Normalization)

    conv2d_3 (Conv2D)          (None, 10, 10, 1024)      745472
['add_10[0][0]']

    block13_pool (MaxPooling2D (None, 10, 10, 1024)      0
['block13_sepconv2_bn[0][0]']
    )

    batch_normalization_3 (Bat (None, 10, 10, 1024)      4096
['conv2d_3[0][0]']
    chNormalization)

    add_11 (Add)                (None, 10, 10, 1024)      0
['block13_pool[0][0]',
'batch_normalization_3[0][0]']

]

    block14_sepconv1 (Separabl (None, 10, 10, 1536)      1582080
['add_11[0][0]']
    eConv2D)

    block14_sepconv1_bn (Batch (None, 10, 10, 1536)      6144
['block14_sepconv1[0][0]']
    Normalization)

    block14_sepconv1_act (Acti (None, 10, 10, 1536)      0
['block14_sepconv1_bn[0][0]']
    vation)

    block14_sepconv2 (Separabl (None, 10, 10, 2048)      3159552
['block14_sepconv1_act[0][0]']
    eConv2D)

    block14_sepconv2_bn (Batch (None, 10, 10, 2048)      8192
['block14_sepconv2[0][0]']
    Normalization)

    block14_sepconv2_act (Acti (None, 10, 10, 2048)      0
['block14_sepconv2_bn[0][0]']
    vation)

    global_average_pooling2d ( (None, 2048)              0

```

```

['block14_sepconv2_act[0][0]']
GlobalAveragePooling2D)

dense (Dense)                (None, 128)                262272
['global_average_pooling2d[0]
                                0]']

dropout (Dropout)            (None, 128)                0
['dense[0][0]']

dense_1 (Dense)               (None, 1)                  129
['dropout[0][0]']

```

```

=====
Total params: 21123881 (80.58 MB)
Trainable params: 262401 (1.00 MB)
Non-trainable params: 20861480 (79.58 MB)
-----

```

```

[16]: # Define the number of epochs
epochs = 10 # You can adjust this number based on your experiments

# Train the model using the training and validation generators
history = model.fit(
    train_generator,
    validation_data=validation_generator,
    epochs=epochs
)

```

```

Epoch 1/10
265/265 [=====] - 531s 2s/step - loss: 0.6226 -
accuracy: 0.6549 - val_loss: 0.7784 - val_accuracy: 0.4612
Epoch 2/10
265/265 [=====] - 519s 2s/step - loss: 0.5538 -
accuracy: 0.7207 - val_loss: 0.8532 - val_accuracy: 0.4806
Epoch 3/10
265/265 [=====] - 518s 2s/step - loss: 0.5160 -
accuracy: 0.7470 - val_loss: 0.8831 - val_accuracy: 0.4891
Epoch 4/10
265/265 [=====] - 512s 2s/step - loss: 0.4963 -
accuracy: 0.7548 - val_loss: 0.9067 - val_accuracy: 0.4749
Epoch 5/10
265/265 [=====] - 614s 2s/step - loss: 0.4824 -
accuracy: 0.7632 - val_loss: 0.9822 - val_accuracy: 0.4882
Epoch 6/10
265/265 [=====] - 666s 3s/step - loss: 0.4752 -

```

```

accuracy: 0.7665 - val_loss: 1.0015 - val_accuracy: 0.4688
Epoch 7/10
265/265 [=====] - 504s 2s/step - loss: 0.4636 -
accuracy: 0.7680 - val_loss: 1.0202 - val_accuracy: 0.4744
Epoch 8/10
265/265 [=====] - 479s 2s/step - loss: 0.4556 -
accuracy: 0.7764 - val_loss: 1.0484 - val_accuracy: 0.4735
Epoch 9/10
265/265 [=====] - 479s 2s/step - loss: 0.4507 -
accuracy: 0.7776 - val_loss: 1.0287 - val_accuracy: 0.4754
Epoch 10/10
265/265 [=====] - 475s 2s/step - loss: 0.4461 -
accuracy: 0.7801 - val_loss: 1.0756 - val_accuracy: 0.4886

```

```

[17]: # Evaluate the model on the validation data
val_loss, val_accuracy = model.evaluate(validation_generator)
print(f"Validation Loss: {val_loss:.4f}")
print(f"Validation Accuracy: {val_accuracy:.4f}")

```

```

66/66 [=====] - 94s 1s/step - loss: 1.0906 - accuracy:
0.4858
Validation Loss: 1.0906
Validation Accuracy: 0.4858

```

```

[20]: from sklearn.metrics import confusion_matrix, classification_report

# Reset the validation generator to ensure predictions start from the beginning
validation_generator.reset()

# Get predictions on the validation set
preds = model.predict(validation_generator)
predicted_classes = (preds > 0.5).astype(int).ravel()

# Get true class labels
true_classes = validation_generator.classes
class_labels = list(validation_generator.class_indices.keys())

# Generate and print the confusion matrix
cm = confusion_matrix(true_classes, predicted_classes)
print("Confusion Matrix:")
print(cm)

# Generate and print the classification report
report = classification_report(true_classes, predicted_classes,
    ↪target_names=class_labels)
print("Classification Report:")
print(report)

```

```

66/66 [=====] - 94s 1s/step

```

Confusion Matrix:

```
[[776 190]
 [901 245]]
```

Classification Report:

	precision	recall	f1-score	support
fake	0.46	0.80	0.59	966
real	0.56	0.21	0.31	1146
accuracy			0.48	2112
macro avg	0.51	0.51	0.45	2112
weighted avg	0.52	0.48	0.44	2112

```
[22]: import matplotlib.pyplot as plt

# Retrieve accuracy and loss values from the training history
acc = history.history['accuracy']
val_acc = history.history['val_accuracy']
loss = history.history['loss']
val_loss = history.history['val_loss']
epochs_range = range(epochs)

# Plot Accuracy
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(epochs_range, acc, label='Training Accuracy')
plt.plot(epochs_range, val_acc, label='Validation Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.title('Training and Validation Accuracy')
plt.legend(loc='lower right')

# Plot Loss
plt.subplot(1, 2, 2)
plt.plot(epochs_range, loss, label='Training Loss')
plt.plot(epochs_range, val_loss, label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Training and Validation Loss')
plt.legend(loc='upper right')
plt.show()
```

